Supplementary material for: A factorised physics prior for Boussinesq-trained surrogates: the compressible correction in buoyancy-driven enclosure convection is linear to leading order in the relative temperature contrast

Abstract

This supplement collects the material that supports the main text but is not needed in the flow of its argument: the asymptotic predictions and their numerical confirmation, the two-parameter fits, the exact single-ε transfer matrix, the learning curve, the public-data splits, the prefactor–velocity law, the pressure-alignment protocol, the shape-stability and data-set tables, the wall-heat-transfer tables, the full three-dimensional fits, and the mesh- and noise-audit details. Table and section numbers in cross-references refer to the main text.

The runs behind every number in the main text are: 600 main solutions (run_batch.sh), 100 paired (run_paired.sh), 27 at $\varepsilon \to 0$ (run_eps0.sh), 104 at $Pr = 4.38$ (run_water.sh); run_cpu_queue.sh adds the 2:1 geometry and the tight-tolerance reruns.

Table S1: The five testable consequences of the expansion and their measured values.

prediction	measurement
$R \to 0$ as $\varepsilon \to 0$	intercept $+0.0003 \pm 0.0047$ (u_x); 0.06σ from zero
$R \propto \varepsilon$	$m = 1.023 \pm 0.018$ / 0.995 ± 0.019 / 1.009 ± 0.015 ($u_x/u_y/p$), window-averaged over the two decades; the residual is the ε^2 term (Sec. 5.1)
the shape of D does not rotate with ε	$\cos(D(\varepsilon), D(1.2)) = 0.95$–$1.00$, four channels × five values of ε
$\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$	$\mathrm{div}(Y)$ vs $\varepsilon \nabla \cdot (\theta \boldsymbol{u}_X)$: cosine $+0.994$ for $\varepsilon < 0.6$, $+0.965$ overall, amplitudes 2.26×10^{-2} vs 2.10×10^{-2}
the expansion converges fast	per configuration, ε^1 alone carries 86% of the correction energy, $\varepsilon^1 + \varepsilon^2$ carries 99.0%, ε^3 carries 99.9%

Table S2: Two-parameter refinement, $R = A\varepsilon^m Ra^q$. The Ra trend for u_y and θ reproduces in the 2:1 geometry.

channel	unit cavity ($n = 600$) m / q / R^2	aspect 2:1 ($n = 104$) m / q / R^2
u_x	1.081 / $+0.003 \pm 0.006$ / 0.74	1.071 / -0.143 ± 0.019 / 0.71
u_y	1.016 / -0.083 ± 0.005 / 0.77	1.146 / -0.077 ± 0.019 / 0.69
θ	1.120 / $+0.514 \pm 0.015$ / 0.72	0.989 / $+0.447 \pm 0.045$ / 0.55
p	1.127 / -0.029 ± 0.005 / 0.83	1.057 / -0.154 ± 0.017 / 0.74

Table S3: Exact transfer error $R(u_x)$ when a field computed at one ε (rows) is applied at another (columns).

$R(u_x)$ taken from ↓ / applied at →	0.4	0.8	1.2	1.6	1.9
$\varepsilon = 0.4$	0	0.067	0.143	0.235	0.353
$\varepsilon = 0.8$	0.068	0	0.078	0.183	0.305
$\varepsilon = 1.2$	0.149	0.081	0	0.110	0.243
$\varepsilon = 1.6$	0.265	0.199	0.112	0	0.140
$\varepsilon = 1.9$	0.402	0.328	0.251	0.142	0

Table S4: Learning curve on the fixed interpolation test set (200 samples), mean of three seeds. Entries are $u_x/u_y/\theta/p$. P3 keeps ε outside the network entirely.

n	B: ε as input	P3: predict $P(X)$, ε outside
0 (do nothing)	0.2203/0.2683/0.0381/0.2386	same
50	0.1831/0.1946/0.0269/0.1838	0.1308/0.1364/0.0195/0.1501
100	0.1280/0.1422/0.0199/0.1348	0.0990/0.1022/0.0136/0.1150
200	0.0974/0.1070/0.0147/0.0998	0.0827/0.0866/0.0114/0.1005
400	0.0674/0.0714/0.0097/0.0672	0.0735/0.0743/0.0093/0.0896

Table S5: Median relative correction on all five public splits, at $\varepsilon \equiv 1.2$.

split	n	u_x	u_y	θ	p
train (random walls)	4000	0.1797	0.2435	0.0228	0.1858
test_in_distribution	200	0.1822	0.2388	0.0220	0.1863
test_ood (harder wall dials)	200	0.1567	0.2157	0.0214	0.1671
train_legacy (classical cavity)	5000	0.2988	0.3362	0.0926	0.1989
test_legacy (classical cavity)	200	0.2984	0.3390	0.0836	0.1983

Table S6: The θ-channel prefactor is the flow strength: $R_\theta/\varepsilon = c\hat{u}^a$ on six data sets from two independent generators.

data set	n	$R_\theta/\varepsilon = c\hat{u}^a$	R^2
ours, random walls, $Pr = 0.71$	600	$a = 1.864 \pm 0.026$, $c = 0.544$	0.90
public train, random walls	4000	$a = 1.902 \pm 0.010$, $c = 0.602$	0.91
public test_in_distribution	200	$a = 1.977 \pm 0.041$, $c = 0.691$	0.92
public test_ood	200	$a = 1.931 \pm 0.041$, $c = 0.561$	0.92
public train_legacy, classical cavity	5000	$a = 2.360 \pm 0.002$, $c = 2.063$	0.995
public test_legacy, classical cavity	200	$a = 2.368 \pm 0.011$, $c = 2.095$	0.996

S1 A POD-ROM check of the prior

The fifteen samples at $Ra = 10^4$ (described in Sec. S9) also support a linear surrogate check. We train a POD-ROM with at most four modes on three flows and four contrasts (twelve samples) and test on the held-out contrast (three samples), repeating for each of the five contrasts. The plain ROM predicts D from $[1, \varepsilon, \boldsymbol{c}_X]$, where $\boldsymbol{c}_X$ are at most four POD coefficients of the Boussinesq field; the prior ROM predicts $P = D/\varepsilon$ from $[1, \boldsymbol{c}_X]$ and multiplies by ε outside. At most four modes are used, fewer than the twelve training samples. Averaged over the five held-out contrasts, the prior lowers the relative error in every channel (Table S8): u_x from 0.168 to 0.081, u_y from 0.190 to 0.150, θ from 0.034 to 0.027 and p from 0.163 to 0.098. The gain is largest at $\varepsilon = 0.2$ and 0.4 (14× and 5× in u_x) and is not uniform: at $\varepsilon = 1.2$ the ε-aware plain ROM is better in all four channels. We read this as a check that the factorisation remains useful in three dimensions at small contrast, not as a claim of uniform superiority.

S2 The Prandtl-number pair

Same configuration, same ε, only Pr changed: 104 paired samples at $Pr = 0.71$ and $Pr = 4.38$. The table gives the fits, the figure the paired comparison.

S3 Cost and speed-up

Wall-clock timings behind the speed-up numbers in the main text.

S4 The pressure channel: raw and aligned statistics

The two members store pressure in different conventions. The compressible raw pressure carries a hydrostatic ramp 12× larger than its dynamic part, and the two weak forms use opposite sign conventions. Comparing

Table S7: Scripts and the results they produce.

script	what it produces
exp_eps_law.py	the 627-sample law (Sec. 5.1)
exp_grid128.py	the 64^2 vs 128^2 mesh check at $Ra = 10^4$ and $Ra = 10^6$ (Sec. 5.1)
repack_p_fixed.py	the corrected pressure channel (Sec. 5.2)
exp_paired_final.py	the ε-blind floor and shape invariance (Sec. 5.3)
exp_modes.py	the ε^k mode expansion (Sec. 3)
exp_chi.py	the χ explanation for θ (Sec. 5.1)
harness.py, exp_ml_final.py	the unified surrogate ablation: A/B/P1/P2/P3, three seeds (Secs. 5.4–5.6)
fno_lib.py, exp_fno_final.py	the FNO baseline under the same protocol (Sec. 5.5)
exp_lc_final.py	learning curves (Sec. 5.6)
exp_pod.py	the POD-ROM / linear-surrogate baselines (Sec. 5.4)
exp_direct_linear.py, exp_direct_batch.py	the direct solution of the linearised $O(\varepsilon)$ system on all 20 configurations (Sec. 5.4)
exp_direct_eps0.py	the same one-Newton-step solve on the 27 low-contrast samples (Sec. 5.4)
exp_heldout_loweps.py, run_loweps.sh	the 38 held-out solutions at $\varepsilon = 0.1, 0.2$ and the extrapolated-P check (Sec. 5.4)
exp_direct_low.py, exp_fixed_flow_eps.py	one-Newton-step solves at $\varepsilon = 0.1, 0.2$ on the paired flows and the fixed-flow ε-drift (Sec. 5.4)
exp_cluster_boot.py	family-cluster bootstrap of the exponent (Sec. 5.1)
exp_public_transfer.py	external validation and transfer (Secs. 5.7–5.9)
exp_aspect2.py	the 2:1 geometry (Sec. 5.10)
exp_tight.py, verify_noise.py, exp_final_numbers.py	noise control and timing (Secs. 5.11–5.12)
exp_water.py	the strictly paired $Pr = 4.38$ fluid study (Sec. 5.13)
exp_perturbation.py	the first-order perturbation baseline on the ε scan (Sec. 5.4)
exp_linear_response.py	the linear-response and secant baselines from the paired scan (Sec. 5.4)
exp_second_order.py	the two-term $P + \varepsilon Q$ fit from the paired scan (Supplementary Sec. S10)
exp_nu_clean_summary.py	the isothermal-sidewall Nu statistics and the denominator audit (Sec. 5.8)
exp_nu_grid.py, exp_nu_grid_ra1e6.py, exp_nu_grid_ra1e5.py	the 64^2 versus 128^2 wall-flux check (Supplementary Sec. S10)
exp_public_nu.py, exp_flux_loss.py, exp_flux_stencil.py	wall heat transfer of the public-data surrogate, with and without the wall-flux training penalty, and the cross-stencil check (Sec. 5.8)
exp_film.py	the FiLM-conditioned baseline (Sec. 5.5)
exp_3d_law.py, exp_3d_exp.py, exp_2d_ra_window.py, merge_3d.py, exp_3d_summary.py, make_fig9.py, exp_3d_rom_eps.py	the twenty-four-sample three-dimensional check at two Rayleigh numbers and the POD-ROM baseline (Sec. 5.14)
verify_conventions.py	the 38 400-value convention audit against the published relL2 columns (Sec. 5.7)
exp_public_splits.py, exp_legacy5000.py, exp_uhat_law.py	the five public splits and the $\hat{u}$ law (Sec. 5.7)
exp_transfer_uhat.py	the $\hat{u}$ transfer criterion on the public splits (Sec. 5.9)
audit_v2.py	robustness checks: m vs fitting window, κ, pressure-pair conventions, $\hat{u}$ medians, the 2:1 σ distances
make_figs.py, make_figs2.py	every figure, from the JSON results
make_tables.py	regenerates every table from the JSON results

Table S8: Three-dimensional POD-ROM check on the fifteen paired samples at $Ra = 10^4$. Leave-one-contrast-out: train on three flows × four ε (twelve samples), test on the held-out ε (three samples), repeated for the five contrasts. Entries are mean relative L_2 errors over the five folds (mean ± s.d.); the ROM uses at most four POD modes and ridge regression ($\lambda = 10^{-4}$). The gain is the plain-ROM error divided by the prior-ROM error.

channel	do nothing	plain ROM	prior ROM	gain
u_x	0.1934±0.132	0.1676±0.071	0.0807±0.061	2.08
u_y	0.1880±0.127	0.1901±0.046	0.1497±0.100	1.27
θ	0.0362±0.025	0.0338±0.008	0.0270±0.019	1.25
p	0.1617±0.117	0.1634±0.065	0.0980±0.077	1.67

Table S9: The fluid axis: 104 strictly paired configurations at $Pr = 0.71$ and $Pr = 4.38$ (same seed stream, same Ra and ε). Both columns are fitted on the same 104 configurations over their full $\varepsilon \in [0.21, 1.89]$ range; the exponents are steeper than the full-window values of Table 3 because the sample subset differs, and the two are not directly comparable. The exponent agrees between the two fluids within 1.5%, the paired median ratio of R is 0.95–1.01, and the fitted prefactors move by 3.3% (u_x), 0.2% (u_y), 1.2% (θ) and 6.1% (p).

channel	A, $Pr = 0.71$	A, $Pr = 4.38$	m, $Pr = 0.71$	m, $Pr = 4.38$
u_x	0.1448	0.1496	1.157	1.150
u_y	0.1981	0.1985	1.047	1.049
θ	0.0081	0.0082	1.668	1.686
p	0.1519	0.1426	1.109	1.124

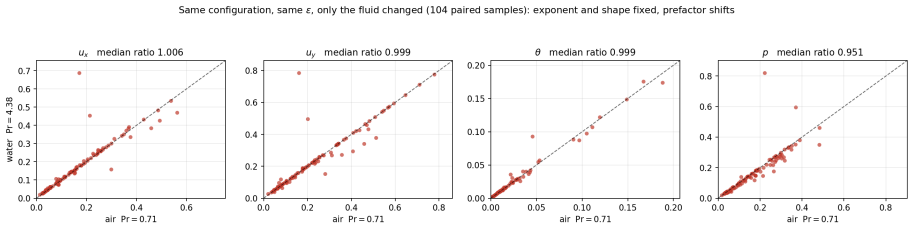


Figure S1: Same configuration, same ε, only the Prandtl number changed: $Pr = 0.71$ against $Pr = 4.38$, 104 paired samples. The panel labels follow the original run names; both runs use the same equation of state and the same dimensionless transport model; only Pr (and hence the thermal diffusivity at fixed Ra) differs.

Table S10: Wall-clock cost, 64^2 mesh. Solver rows use eight CPU threads; the surrogate runs on an RTX 4060 Laptop GPU (batch 1) or one CPU thread. The two members are timed separately over the 600 samples; the rows below them are surrogate timings or totals derived from the measured medians. The 100-sample scan rows use that scan's own timings: the component medians are 16.0 s (Boussinesq) and 37.5 s (compressible), while the median of their per-pair sum is 55.8 s; medians are not additive.

quantity	median	mean	p90	max
paired solve, 600 samples	33.9 s	38.0 s	59.9 s	510 s
Boussinesq member alone	11.7 s	13.1 s	21.7 s	—
compressible member alone	22.3 s	25.0 s	37.3 s	—
paired solve, 100-sample ε scan (per sample)	55.8 s	67.7 s	136 s	202 s
surrogate, GPU, batch 1	6.3 ms			
surrogate, GPU, batch 256	0.49 ms/sample			
surrogate, CPU, 1 thread	14.0 ms			
end-to-end pair (Boussinesq + surrogate)	11.7 s			
one flow, 100 contrasts (scan timing)	16.6 s			
20-flow × 5-contrast scan, surrogate total	321 s			

raw pressures therefore measures the conventions rather than the flow: the per-sample median of $|X_p|/|Y_p|$ is 0.065, that of $\cos(X_p, Y_p)$ is -0.105, and the p-channel "law" collapses to a constant ($R_p \approx 1.0$). We align the members with the protocol used by the public data set—subtract the mean, remove a least-squares linear-in-y fit, and flip the sign of the Boussinesq member. Under the same median convention the aligned pair gives 0.960 and cos = +0.991, against 0.929 and +0.986 for the public training set, so the two pairings agree to within 3.5%. The channel then follows the same law as the velocities ($A = 0.148$, $m = 1.009$). All p-channel numbers in the paper use this gauge. Pressure is gauge-dependent: the exponent is a statement about this fixed post-processing gauge, not a gauge-invariant law.

S5 Shape stability of the correction

The amplitude changes with contrast; the shape does not. Over the 20 paired configurations the median cosine similarity between $D(\varepsilon)$ and $D(1.2)$ stays above 0.952 for $\varepsilon = 0.4$–1.9, so a scalar amplitude carries the contrast dependence.

S6 Data sets

The table lists the sets used in the main text.

Table S11: Cosine similarity between the correction field $D(\varepsilon)$ and $D(1.2)$, median over the 20 paired configurations. Values are ordered by ε.

channel	0.4	0.8	1.2	1.6	1.9
u_x	0.978	0.994	1.000	0.992	0.974
u_y	0.993	0.998	1.000	0.996	0.976
θ	0.991	0.997	1.000	0.996	0.982
p	0.964	0.991	1.000	0.990	0.952

Table S12: Data sets used in this study. The public sets are from Menglik et al. [5].

set	n	what varies	purpose
random-BC	600	$Ra \sim \log U[10^2, 10^6]$, $\varepsilon \sim U[0.2, 1.9]$, 134 wall-mode families	calibrate the law, train surrogates
$\varepsilon \to 0$ checks	27	$\varepsilon = 0.02/0.05/0.10 \times Ra = 10^3/10^4/10^5$, three draws each	anchor $R(0) = 0$ (included in the fit)
paired	20×5	only ε; Ra and the θ boundary stream fixed within a configuration	ε-blind reference, shape invariance
aspect 2:1	104	same law, geometry 2×1	external validity
tight rerun	10	the 10 worst-converged samples, tolerance $10^{-6} \to 10^{-8}$	noise control
public, random walls [5]	$4000 + 2 \times 200$	$\varepsilon \equiv 1.2$; two held-out splits differ in the range of the wall dials	external validation, cross-data-set transfer
public, classical cavity [5]	$5000 + 200$	constant hot/cold side walls, adiabatic top and bottom, $\varepsilon \equiv 1.2$; its 200 held-out pairs are the third split above	external validation of the prefactor
$Pr = 4.38$ (this work)	104	$Pr = 4.38$; BC/Ra/ε draws bit-identical to 104 of the random-BC samples	test of the fluid axis

S7 Wall heat transfer: pair statistics and the Rayleigh trend

The first table gives the pair statistics at $\varepsilon = 1.2$; the second gives the integral change for the five isothermal-sidewall configurations, one at each Rayleigh number.

Table S13: Wall heat transfer at $\varepsilon = 1.2$ over the 200 classical-cavity samples of the public data set. The local change is $\Delta Nu(y)/Nu(y)$ along the hot wall. The integral change is the mean of the per-sample ratios at the six-point stencil, with the median and the 10–90% range; the ratio of the ensemble means is +0.67%. The stencil range is the two- to eight-point one-sided estimate of the wall gradient.

quantity	value	spread over samples
Nu, Boussinesq X	3.514	—
Nu, compressible Y	3.538	—
integral change $\Delta Nu/Nu$	+0.54%	median +0.58%, p10–p90 +0.06 to +0.92%; per-sample mean +0.35 to +0.62% over stencils
local change, L_1 norm over the wall	4.25%	p90 7.33%; unchanged for two to eight points
local change, pointwise	−0.65% (median)	p10–p90 −9.1 to +9.8%

Table S14: Integral Nusselt change $\Delta Nu/Nu$ (percent) for the five isothermal-sidewall configurations, one at each Ra. These are the only configurations with a positive $O(1)$ denominator at every contrast; this is the classical-cavity family, fixed before the analysis. Random-wall families are excluded because 8 of 15 have $|Nu_X| \leq 1$ at $\varepsilon = 1.9$. At $Ra = 10^2$ the entries are below the 0.35% estimator error and are listed only for completeness. At $Ra = 10^6$ the $\varepsilon = 1.2$ entry falls to +0.44% on a 256^2 mesh and the $\varepsilon = 1.9$ entry to −2.10% on a 192^2 mesh.

Ra	$\varepsilon = 0.4$	0.8	1.2	1.6	1.9
10^2	−0.00	+0.00	+0.01	+0.01	−0.01
10^3	+0.07	+0.28	+0.59	+0.76	−0.98
10^4	+0.07	+0.20	+0.27	−0.38	−5.14
10^5	+0.16	+0.48	+0.87	+0.91	−1.81
10^6	+0.24	+0.51	+0.82	+0.83	−1.50

S8 Public-data splits and the theta collapse

The release publishes three held-out splits generated by the same generator and the same solver commit (v6_freefall_20260921, raw_both_deramp pressure convention; the OOD split carries the later tag v6_freefall_ood_20260925): test_in_distribution (200 random-wall pairs), test_ood (200 random-wall pairs whose vertical-wall Fourier dials lie outside the training range) and test_legacy (200 classical-cavity pairs). The release also contains train_legacy, 5000 classical-cavity pairs, used in the transfer experiment of

the main text. The θ-channel prefactor collapses on the peak Boussinesq speed $\hat{u}$ across two independent generators and two boundary-condition families.

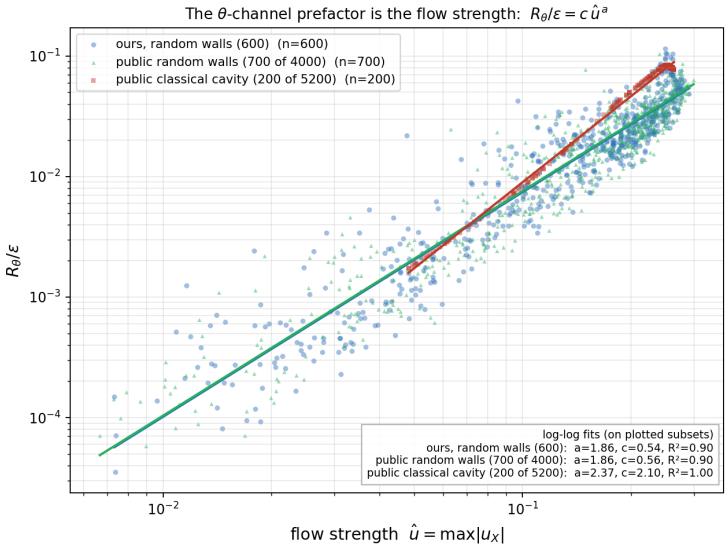


Figure S2: The θ-channel prefactor collapses on the peak Boussinesq speed $\hat{u}$ across two independent generators and two boundary-condition families.

S9 A three-dimensional check

The law is calibrated in two dimensions. To check whether the exponent survives in three, we generated twenty-four paired steady solutions with the same generator: three wall types—a three-cell Voronoi pair, a two-cell Voronoi pair and a constant-wall cavity—at $Ra = 10^4$ over $\varepsilon = 0.2$, 0.4, 0.8, 1.2 and 1.6 (fifteen samples), and the same three wall types at $Ra = 3 \times 10^4$ over $\varepsilon = 0.4$, 1.2 and 1.9 (nine samples), with $Pr = 0.71$ and $res = 16$ (33^3 velocity nodes, 17^3 temperature and pressure nodes). Within each set the pairing is exact: the wall temperatures are the same seed stream scaled affinely about T_0, so the low-fidelity member is the same solution up to the ε-blind rescaling of Sec. 2.3. For the pressure channel we use the same fixed gauge as in two dimensions: subtract the mean, remove a linear fit in y (the gravity direction), and flip the sign; the horizontal z direction is not deramped. The p exponents below are therefore fixed-gauge numbers, as in Table 3.

Table S15 and Fig. S3 give the result. The fit is the one the two-dimensional law takes, $R_{f,c} = A_{f,c}\varepsilon^{m_c}$: a separate prefactor for each flow and channel, and one exponent per channel and Rayleigh number. At $Ra = 10^4$ the exponents for u_x, u_y and p are 1.091±0.013, 1.052±0.011 and 1.141±0.020

($R^2 \geq 0.997$); at $Ra = 3 \times 10^4$ they are 1.026 ± 0.066, 1.041 ± 0.063 and 1.086 ± 0.110 ($R^2 \geq 0.96$). All six lie within 0.15 of unity. The largest difference between the two Rayleigh numbers is 0.065 (u_x), against a combined standard error of 0.067; the comparison is inconclusive and does not exclude a Rayleigh dependence of about 0.2 over a factor of three. The θ rows, 1.130 ± 0.019 and 1.041 ± 0.081 (difference 0.089 ± 0.083), are reported separately; θ does not enter the common-exponent claim here any more than in Sec. 5.1.

The two sets also cover different windows, 0.2–1.6 and 0.4–1.9. Refitting the $Ra = 10^4$ data on the three contrasts $\varepsilon = 0.4$, 1.2 and 1.6 (nine samples) gives 1.126, 1.074 and 1.194 for u_x, u_y and p (θ: 1.179), with standard errors of 0.02–0.03; the three velocity and pressure values are then 0.03–0.11 above the $Ra = 3 \times 10^4$ values on the same lower window (the windows differ only at the top end, 1.6 against 1.9). These are window-local exponents over $\varepsilon \leq 1.6$, so the two-dimensional comparison must use the same window and a comparable Rayleigh number. Fitting the 2-D samples in the Ra bins 5×10^3–2×10^4 and 2×10^4–6×10^4 over $\varepsilon = 0.4$–1.9 gives 1.113 ± 0.091, 1.135 ± 0.076, 1.164 ± 0.209 and 1.200 ± 0.068 at $Ra \approx 10^4$ (77 samples) and 1.185 ± 0.096, 1.178 ± 0.086, 1.222 ± 0.208 and 1.087 ± 0.075 at $Ra \approx 3 \times 10^4$ (79 samples) for $u_x/u_y/\theta/p$. The 3-D values agree with these within 1.4 combined standard errors. The two-decade fits of Table 3 average over the curvature and are not the right comparison. The prefactor is not universal either: across the three flows it varies by 3.9× (u_x), 2.1× (u_y), 10.4× (θ) and 2.0× (p) at $Ra = 10^4$, and by 3.7×, 2.7×, 13.1× and 2.4× at $Ra = 3 \times 10^4$. The 3-D runs are therefore consistent with the same near-unity, window-local exponent for u_x, u_y and p; θ is reported separately. The amplitude remains a functional of the flow. With one resolution and two Rayleigh numbers, this is a consistency check, not a converged three-dimensional validation.

A POD-ROM check of the prior. The fifteen samples at $Ra = 10^4$ also support a linear surrogate check. We train a POD-ROM with at most four modes on three flows and four contrasts (twelve samples) and test on the held-out contrast (three samples), repeating for each of the five contrasts. The plain ROM predicts D from $[1, \varepsilon, \boldsymbol{c}_X]$, where $\boldsymbol{c}_X$ are at most four POD coefficients of the Boussinesq field; the prior ROM predicts $P = D/\varepsilon$ from $[1, \boldsymbol{c}_X]$ and multiplies by ε outside. At most four modes are used, fewer than the twelve training samples. Averaged over the five held-out contrasts, the prior lowers the relative error in every channel (Table S8): u_x from 0.168 to 0.081, u_y from 0.190 to 0.150, θ from 0.034 to 0.027 and p from 0.163 to 0.098. The gain is largest at $\varepsilon = 0.2$ and 0.4 (14× and 5× in u_x) and is not uniform: at $\varepsilon = 1.2$ the ε-aware plain ROM is better in all four channels. We read this as a check that the factorisation remains useful in three dimensions at small contrast, not as a claim of uniform superiority.

The check is limited to two Rayleigh numbers, $res = 16$ and $\varepsilon \leq 1.9$.

The twenty-four pairs cost 12–32 min each of outer wall-clock time on this machine. We therefore report the exponent transfer, not a three-dimensional sweep.

Table S15: Three-dimensional check: twenty-four strictly paired samples at two Rayleigh numbers ($Pr = 0.71$, $res = 16$). The fit is $R_{f,c} = A_{f,c}\varepsilon^{m_c}$, with one prefactor per flow and channel and one exponent per channel and Rayleigh number; each row is a separate fit. Flows are ordered three-cell Voronoi, two-cell Voronoi, constant wall, and the prefactor is listed per flow to show that it does not transfer between them. The fit is unweighted least squares in log space with three flow intercepts; $n = 15$ at $Ra = 10^4$ and $n = 9$ at $Ra = 3\times10^4$ (11 and 5 residual degrees of freedom); ± is the standard error of m and R^2 is computed in log space. θ is wall-normalised, $(T - T_c)/\Delta T$, and is not part of the common-exponent claim.

channel	Ra	m	A per flow	R^2
u_x	10^4	1.091 ± 0.013	0.0670/0.1862/0.2609	0.999
u_y	10^4	1.052 ± 0.011	0.1230/0.1614/0.2528	0.999
θ	10^4	1.130 ± 0.019	0.0064/0.0243/0.0666	0.999
p	10^4	1.141 ± 0.020	0.0972/0.1844/0.1939	0.997
u_x	3×10^4	1.026 ± 0.066	0.0767/0.0975/0.2809	0.988
u_y	3×10^4	1.041 ± 0.063	0.0922/0.0921/0.2529	0.988
θ	3×10^4	1.041 ± 0.081	0.0088/0.0064/0.0839	0.992
p	3×10^4	1.086 ± 0.110	0.0795/0.1451/0.1942	0.962

S10 Mesh independence, solver noise and the two-term fit

Mesh independence

At $Ra = 10^5$ the same isothermal-wall check at $\varepsilon = 1.2$ and 1.9 gives a wall-Nu change of −0.15 and −0.36 percentage points between 64^2 and 128^2 (Nu_X itself moves by 0.86%); the integral correction moves from +0.87% to +0.73% at $\varepsilon = 1.2$ and from −1.81% to −2.16% at $\varepsilon = 1.9$. Across $Ra = 10^4$, 10^5 and 10^6 the change between 64^2 and 128^2 is therefore at most 0.38 percentage points, and the sign and order of the integral correction are unchanged.

At $Ra = 10^6$ the isothermal-wall pair was refined to 192^2 and, at $\varepsilon = 1.2$, to 256^2. At $\varepsilon = 1.2$ the wall-Nu correction falls from +0.82% at 64^2 to +0.56% at 128^2 and +0.44% at 256^2, where eight temperature nodes span the thermal boundary layer. At $\varepsilon = 1.9$ it moves from −1.50% at 64^2 to −1.88% at 128^2 and −2.10% at 192^2. Both sequences are still moving at the finest mesh, by 0.12 and 0.22 percentage points per step, so the integral values carry a resolution uncertainty of that order; the sign and the order of magnitude do not change. The $\varepsilon = 1.9$ pair at 256^2 did not reach the steady tolerance: at pseudo-time step 0.05 it ran for 4.5 h and stopped at a relative change of 4×10^{-6} per step against the 10^{-6} target, and the step 0.025 retry did not converge either.

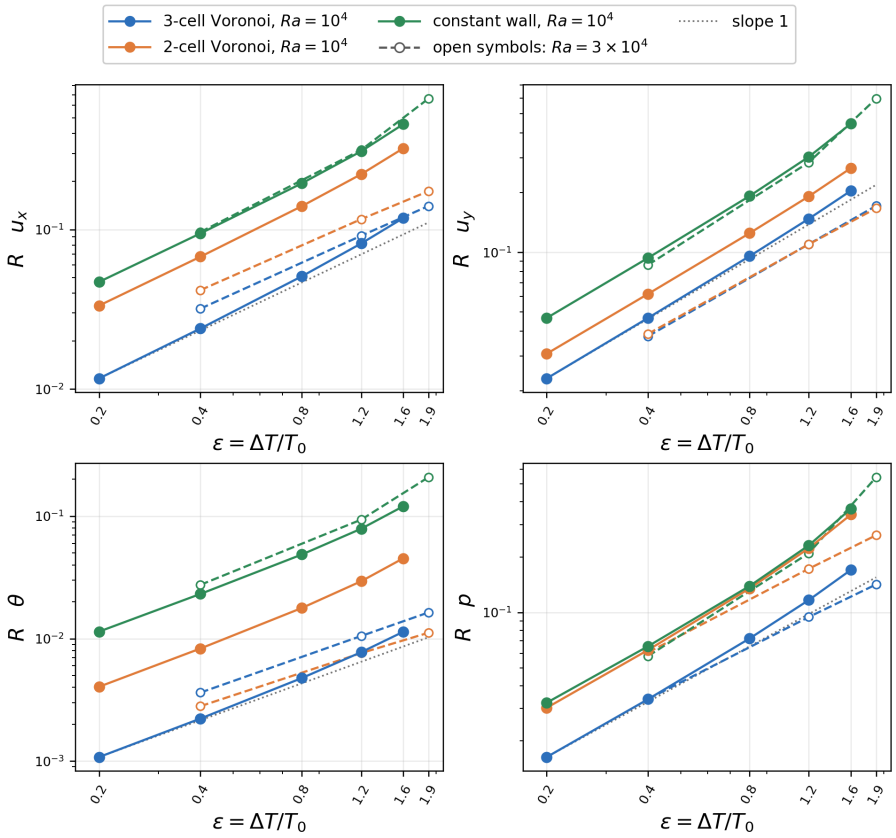


Figure S3: The three-dimensional check in log–log coordinates: twenty-four paired samples, three wall types at two Rayleigh numbers. The vertical axis is $R = \|Y - X\|/\|Y\|$ in the channel labelled in each panel. Filled symbols and solid lines: $Ra = 10^4$ over five contrasts; open symbols and dashed lines: $Ra = 3{\times}10^4$ over three contrasts. Lines connect the samples, not the fitted laws, and open symbols carry the colour of their wall type. The dotted line is a slope-one reference anchored at the lowest ε of the three-cell $Ra = 10^4$ curve.

Solver noise

- Convergence quality over 600 samples (compressible member): median relative change 3.75×10^{-6}, p90 1.09×10^{-4}, worst 7.04×10^{-4}.
- Dropping the 65 samples with relative change $> 10^{-4}$ changes the linear u_x fit from $0.1809\varepsilon - 0.0212$ to $0.1813\varepsilon - 0.0221$—unchanged. The negative intercept is a property of this linear fit; the power-law fit of Table 3 has an intercept consistent with zero.
- Re-running the 10 worst samples with steady_tol tightened from 10^{-6} to 10^{-8} leaves the solution bit-identical ($|\Delta Y|/|Y| = 0$). The reason is structural: the compressible march terminates on the zero-iteration guard, not on steady_tol, so the residual reported is the fixed-point residual of the discrete solution, not unfinished iteration.

A two-term fit

The same low-ε pairs allow a stronger data-driven baseline: fit $\mathcal{P}(\varepsilon) = P + \varepsilon Q$ on $\varepsilon = 0.4, 0.8, 1.2$ for each configuration and predict $X + \varepsilon P + \varepsilon^2 Q$. At $\varepsilon = 1.6$ this gives 3.3/3.9/0.4/3.8% and at $\varepsilon = 1.9$ 11.4/13.9/1.7/13.8% ($u_x/u_y/\theta/p$), against 14/15/1.4/15% and 24/27/3.0/28% for the first-order response. That is the same order as the network's 9.4/10.6/1.4/11.3%, averaged over the extrapolation split; the two are not evaluated on the same samples. The two-term fit is available only when paired low-ε data exist for the target flow. The public-data problem of Sec. 5.7 has no such pairs at the target contrast; there a single-ε data set is all that exists, and the prior—not a quadratic fit—is what makes it reusable.